SonicVitals: Viva Guide to the Dashboards

Every numbered label on the three screenshots, with what it shows, how it is computed and the theory behind it.

The project in one paragraph

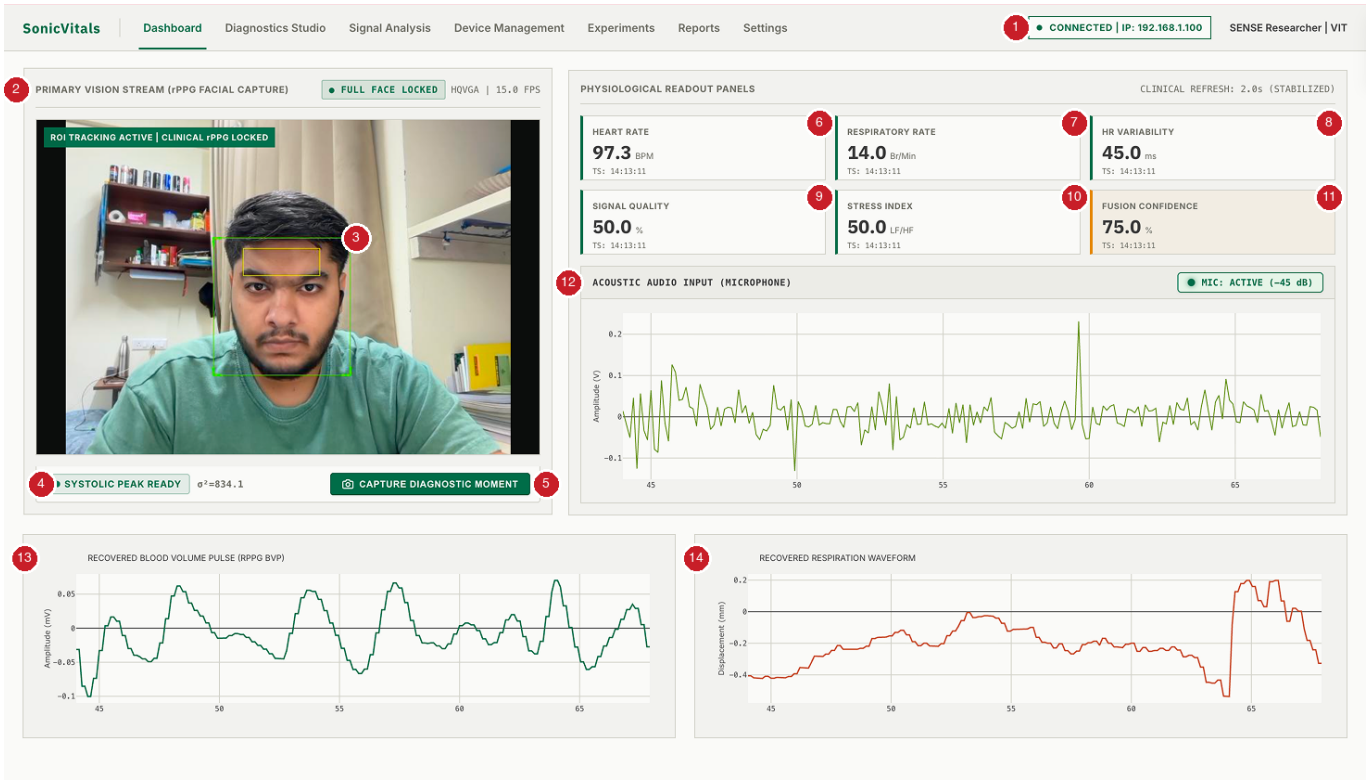
SonicVitals is a contactless vital-signs system. An **ESP32-S3** board carries a camera (GC2145, 240×160 pixels), an **INMP441** I²S MEMS microphone and a 0.96" **SSD1306 OLED**. The board streams JPEG frames and audio to a laptop (over Wi-Fi, TCP port 3333). The laptop runs a Python/Flask server that detects the face, extracts skin colour over time (**remote photoplethysmography, rPPG**) to estimate heart rate (HR) and respiratory rate (RR), and analyses heart sounds (**phonocardiography, PCG**) from the microphone. Results go to the web dashboard and back to the OLED.

Core principle of rPPG. With every heartbeat, more blood enters the skin capillaries. Haemoglobin absorbs light (strongly in green, less in red), so the skin's reflected colour changes by about 0.1–1 % at the heart rate. The eye cannot see it, but the *average* over thousands of skin pixels makes it measurable. **Pipeline:** face detection → skin ROI mean (R, G, B) per frame → CHROM colour projection → band-pass 0.75–2.5 Hz → Welch spectrum → highest peak f_0 → HR = $60 f_0$.

Viva note

These three screenshots come from the **first build** (5 Oct). A few cards in that build showed values produced by fixed formulas or labels, not measurements; they are marked below with "*Viva note*". Say this openly if asked. It shows you understand your own system, and the current build has already fixed them.

1 Dashboard (live monitoring)



1 Connection badge. *What:* shows whether the host is receiving data from the node. *How:* the server polls the link; the badge turns green when frames arrive.

Viva note

"IP 192.168.1.100" was fixed text in the first build's HTML. The current badge shows the real link (e.g. `tcp://sonicvitals.local:3333`).

2 Primary vision stream. *What:* the live camera picture after processing, served by Flask as an MJPEG stream (`/video_feed`). "FULL FACE LOCKED" means the frame passed the five-test face gate (item 3). "HQVGA" is the 240×160 resolution. *Theory:* rPPG needs a still, well-lit, fully visible face; anything else adds noise far bigger than the pulse.

Viva note

"15.0 FPS" was a fixed label. The real frame rate over Wi-Fi is about 8 frames/s (measured; the dashboard now shows the measured value).

3 Face box (green) and forehead ROI (yellow). *What:* where the face was found and which skin patch gives the pulse. *How:* **Viola–Jones Haar cascade** (OpenCV): the image is converted to grey and histogram-equalised, then scanned with rectangular Haar features evaluated quickly through an *integral image*. AdaBoost-selected features are arranged in a *cascade*, so non-face windows are rejected early. The box is smoothed over frames to stop jitter. **Five-test gate**: (1) face not touching the edges, (2) face size 20–88 % of the frame width, (3) aspect ratio 0.65–1.30 (not turned), (4) all ROIs inside the image, (5) enough light (cheek brightness > 15) and similar left/right brightness. Only frames that pass enter the analysis. *Current build:* forehead *plus both cheeks* (more skin pixels means less noise).

4 "Systolic peak ready" and $\sigma^2 = 834.1$. *What:* σ^2 is the **sharpness** of the best frame in the last 8 s: the variance of the Laplacian (2nd-derivative filter) of the grey image. A blurred or moving image has weak edges, so a low variance; higher means sharper. The sharpest frame is kept for the colour analysis (Diagnostics Studio).

Viva note

"Systolic peak" was set when the latest green value was above its 10-frame mean. That is not a reliable systole detector (in systole green *drops*), so it was removed from the current build.

5 Capture Diagnostic Moment. *What:* sends the sharpest gated frame to `/api/capture_diagnostic`, which computes the colour indices shown in the Diagnostics Studio.

6 Heart rate (97.3 BPM). *How:* ROI means R, G, B are buffered with timestamps and resampled to a uniform 15 Hz grid. **CHROM** (de Haan & Jeanne, 2013): each channel is divided by its mean over a short window (removes light intensity), then

$$X = 3R_n - 2G_n, \quad Y = 1.5R_n + G_n - 1.5B_n, \quad S = X - \alpha Y, \quad \alpha = \sigma_X/\sigma_Y.$$

X and Y both contain the motion/specular component, so subtracting αY cancels it and leaves the pulse S. S goes through a 3rd-order **Butterworth band-pass** 0.75–2.5 Hz (45–150 BPM), run forwards and backwards (`filtfilt`, zero phase). A **Welch** spectrum (Signal Analysis, item 7) gives the peak f_0; HR $= 60 f_0$. The display is refreshed every 2 s and smoothed with an exponential moving average (EMA). Normal resting adult range: 60–100 BPM.

7 Respiratory rate (14.0 br/min). *Theory:* breathing slowly modulates the skin signal (changes in venous return, chest/head movement and light angle): *respiratory-induced intensity variation*. *How:* the green channel is band-passed 0.15–0.40 Hz (9–24 breaths/min) and the spectral peak f_r gives RR $= 60 f_r$. Normal adult: 12–20 br/min. *Current build:* RR is shown only if the peak holds ≥ 50 % of the low-frequency power (otherwise "–").

8 HR variability (45.0 ms). *Theory:* HRV is the beat-to-beat variation of the interval between heartbeats (IBI). Standard time-domain measure: RMSSD $= \sqrt{\frac{1}{N-1}\sum(IBI_{i+1} - IBI_i)^2}$; typical resting values 20–60 ms.

Viva note

In the first build this card was *not measured*: it was $55 + 0.4(\text{SQI} - 75)$, and with SQI $= 50$ that gives exactly 45.0. The current build computes real RMSSD from heart-sound beat times (needs ≥ 20 clean intervals), else "–".

9 Signal quality (50.0 %). *What:* how dominant the cardiac peak is. First build: $\min(99, \max(50, 160\, P_{\text{peak}}/P_{\text{band}}))$. **50.0 is the floor**, meaning the peak held ≤ 31 % of band power: a *weak* signal at that moment. *Current build:* spectral concentration Q (power within ± 0.1 Hz of f_0 divided by band power), with no clamp.

10 Stress index (50.0, labelled LF/HF). *Theory:* the real LF/HF index is the ratio of HRV spectral power at 0.04–0.15 Hz (sympathetic + parasympathetic) to 0.15–0.4 Hz (parasympathetic); it needs minutes of accurate beat times.

Viva note

It was not LF/HF: it was $50 - 0.5(\text{HRV} - 45)$, which is 50.0 when HRV is 45.0. Removed in the current build.

11 Fusion confidence (75.0 %). *Theory: sensor fusion* combines the camera HR and the microphone HR, weighted by their quality. *Current build:* HR $= (w_o\,\text{HR}_o + w_a\,\text{HR}_a)/(w_o + w_a)$ if the two agree within 8 BPM; otherwise the higher-quality channel is used and a conflict is flagged.

Viva note

First build: $\max(75, 1.05 \cdot \text{SQI})$, so 75.0 is the floor value, not a measured confidence.

12 Acoustic input (MIC −45 dB). *What:* live microphone waveform. -45 dB $= 20\log_{10}$(peak amplitude), i.e. dB relative to full scale (dBFS); quiet input. Tall spikes are clicks or speech. *Theory:* heart sounds S1 (valves closing at

the start of systole) and S2 (end of systole) lie at 20–150 Hz. *Current build:* the ESP32's INMP441 is used. Audio is band-passed 25–400 Hz and resampled to 1 kHz. The **Shannon energy envelope** $-x^2 \log x^2$ boosts medium-intensity sounds and suppresses noise. The envelope's **autocorrelation** peaks at one beat period τ, so HR $= 60/\tau$.

Viva note

In the first build this was the *laptop/browser* microphone (Web Audio API), and the "V" axis was not calibrated volts. Air-coupled MEMS mics also cut off below about 60 Hz, so heart sounds are weak unless the mic is pressed on the chest.

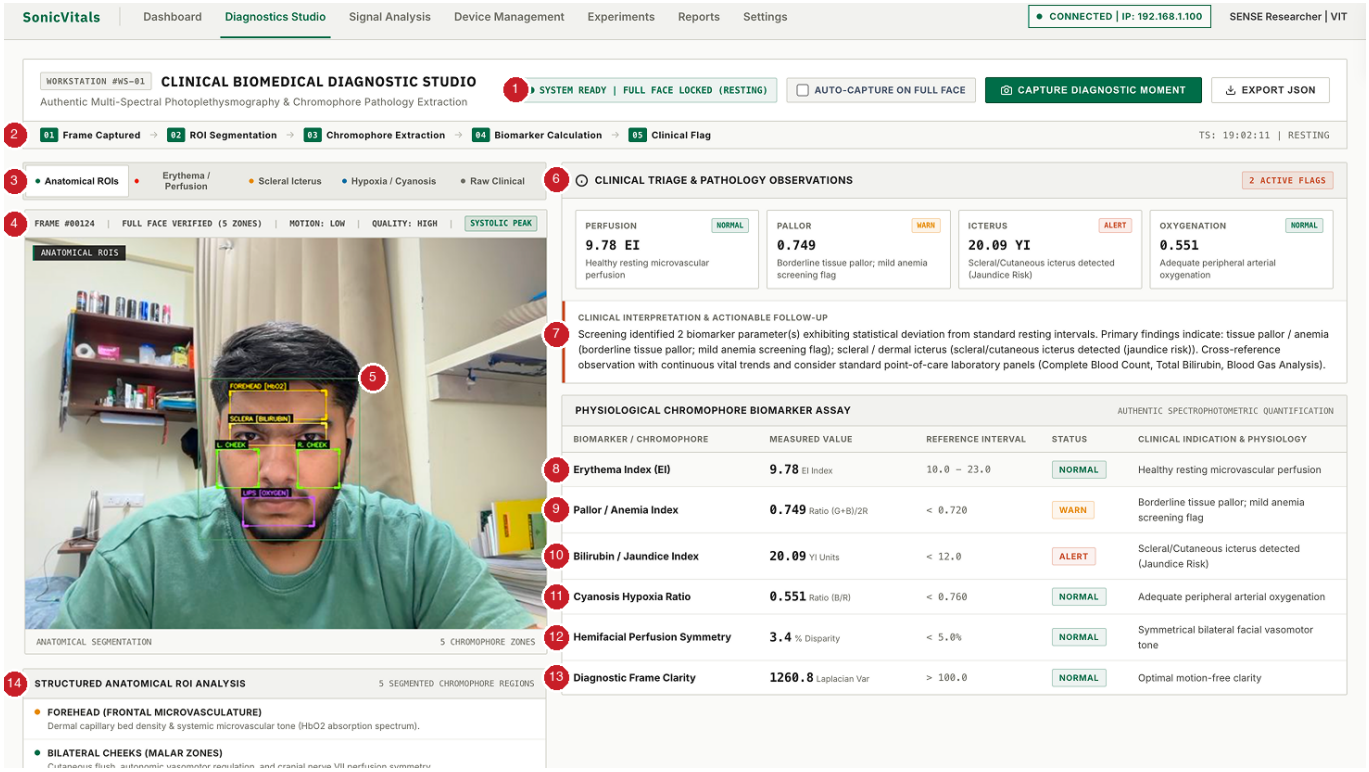
13 **Recovered blood volume pulse (BVP).** *What:* the band-passed CHROM signal, standardised (z-score: mean 0, SD 1). Each hump is one heartbeat (systolic upstroke, then a slower decline). Counting peaks per minute gives HR.

Viva note

Axis units "mV" are cosmetic (the signal is dimensionless). The x-axis advanced 0.12 per dashboard update rather than in real seconds, which is why the humps look slower than 97 BPM. The current build plots against real time.

14 **Recovered respiration waveform.** *What:* the 0.15–0.40 Hz component of the green channel (item 7). One slow cycle ≈ one breath (about 4 s at 14 br/min). The sharp jump near 64 is a **motion artefact** (head movement), which is why quality gating matters. "Displacement (mm)" is a cosmetic label; the signal is a normalised intensity.

2 Diagnostics Studio (colour analysis of one frame)



Idea. Skin colour depends on chromophores: *haemoglobin* (red, perfusion), *deoxy-haemoglobin* (bluish), *bilirubin* (yellow), *melanin* (brown). Ratios of mean R, G, B in chosen face regions give simple **colorimetric indices**. They are **exploratory screening indicators, not diagnoses**: they depend strongly on lighting, camera white balance and skin tone, and were not validated against blood tests.

1 **Status and controls.** Shows that the face gate passed. *Auto-capture* grabs a frame automatically when the gate passes. *Export JSON* saves all values.

2 **Processing steps 01–05.** (01) sharpest gated frame → (02) five ROIs placed relative to the face box → (03) mean R, G, B per ROI → (04) index formulas → (05) compare with software reference ranges (NORMAL / WARN / ALERT).

3 **View tabs.** Switch between the ROI view and false-colour heat maps (OpenCV colour maps) of each index over the face: erythema (R vs. G), yellowness (CIELAB b^*), cyanosis (B/R).

4 **Frame info strip.** Frame number of the captured frame, "5 zones" verified, and *Motion / Quality* labels. *Viva note:* both labels came from the sharpness value only ($> 100 \Rightarrow$ LOW / HIGH); motion itself was not measured.

5 **Five anatomical ROIs.** Boxes at fixed fractions of the face box: **Forehead** (perfusion), **Sclera** band across the eyes (yellowness), **Left/Right cheek** (pallor and symmetry), **Lips** (cyanosis).

6 Triage cards (2 active flags). The four headline indices repeated as cards; "2 active flags" counts rows not NORMAL (Pallor WARN, Icterus ALERT).

7 Interpretation text. Auto-generated sentence listing the flagged indices. *Viva note:* the clinical wording (anemia, jaundice risk, lab panels) has been toned down in the current build to "in/out of software range".

8 Erythema Index, EI $= 9.78$. $\mathrm{EI} = 100\,(\log_{10}\bar{R} - \log_{10}\bar{G})$ on the forehead (Dawson/Diffey style). More blood means more green absorbed, so R/G and EI rise. Here $R/G = 10^{0.0978} \approx 1.25$. 9.78 is NORMAL because the code's actual range was 8–23; the "10.0–23.0" printed in the table was a label mismatch (fixed in the current build).

9 Pallor index $= 0.749$. $\mathrm{PI} = (\bar{G} + \bar{B})/(2\bar{R})$ on both cheeks. Less haemoglobin (anaemia) means less red, so PI rises. $0.749 > 0.72$ gives WARN.

10 Jaundice / yellowness index $= 20.09$. $\mathrm{YI} = 100\,(\bar{R} + \bar{G} - 2\bar{B})/(\bar{R} + \bar{G} + \bar{B})$ on the eye band. Bilirubin absorbs blue, so yellow tissue has low B and high YI. *Viva note:* the band includes skin and eyebrows, and warm room light lowers B, so ALERT here is most likely **lighting/skin tone**, not jaundice. This is the main limitation of RGB colorimetry without a colour reference card.

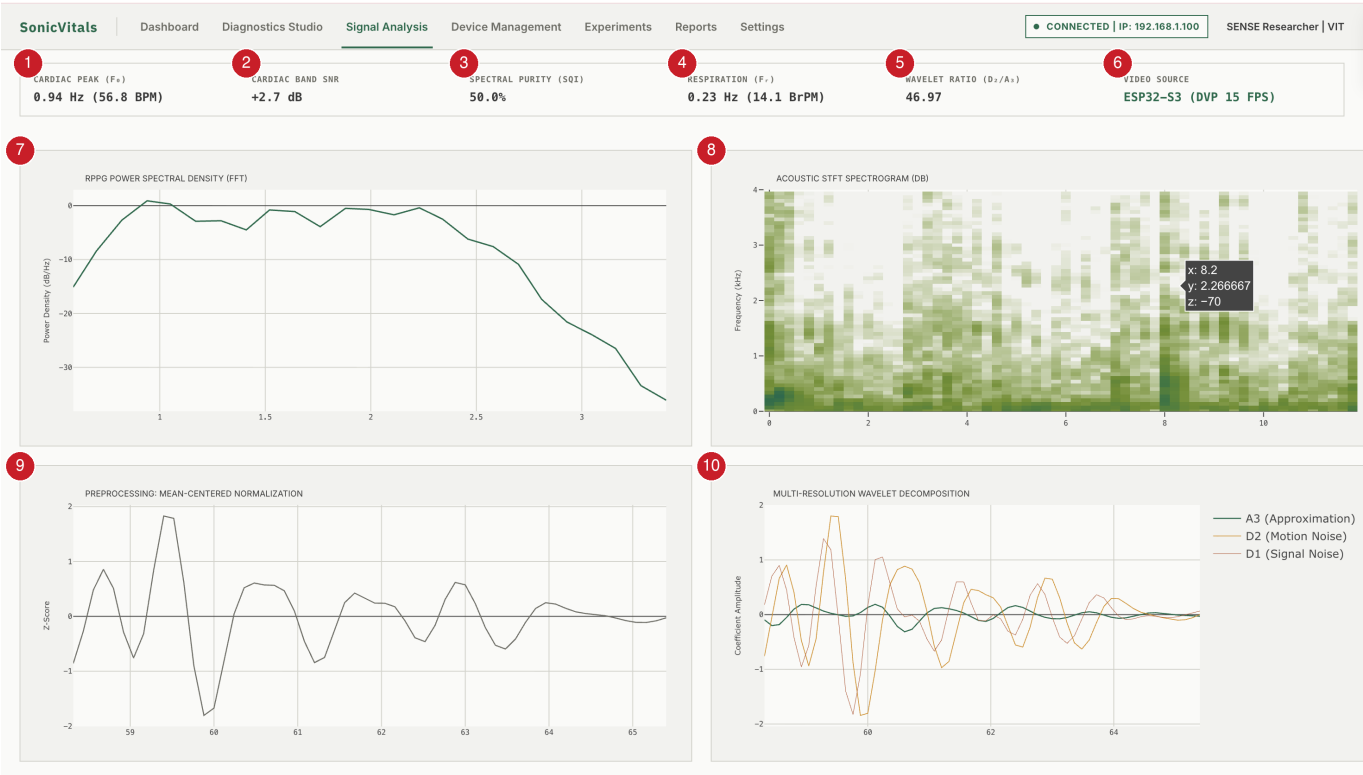
11 Cyanosis ratio $= 0.551$. $\mathrm{CR} = \bar{B}/\bar{R}$ on the lips. Deoxygenated blood looks bluer, so CR rises; below 0.76 is in range.

12 Hemifacial symmetry $= 3.4\,\%$. $100\,|\bar{G}_L - \bar{G}_R|/(\bar{G}_L + \bar{G}_R)$, the left/right cheek difference in green (perfusion). Large asymmetry can also come from side lighting, so it is an observation only.

13 Frame clarity $= 1260.8$. Variance of the Laplacian (see Dashboard item 4); > 100 is accepted as sharp.

14 ROI analysis list. Text description of each region and why it was chosen (e.g. forehead: thin skin, dense capillaries, little hair or motion).

3 Signal Analysis (inside the HR estimate)



1 Cardiac peak $f_0 = 0.94$ Hz (56.8 BPM). Frequency of the highest point of the spectrum (item 7) inside 0.75–2.5 Hz; $\mathrm{HR} = 60 \times f_0$. This screenshot is from a different moment than the Dashboard (97.3 BPM), so the values differ.

2 Cardiac band SNR $= +2.7$ dB. First build: $10\log_{10}(P_{\mathrm{peak}}/\text{median in-band power})$; +2.7 dB means the peak is only 1.9× the typical level, i.e. **weak**. *Current build* uses the standard de Haan SNR: power within ± 0.1 Hz of f_0 and ± 0.2 Hz of $2f_0$ (harmonic) divided by the remaining 0.5–4 Hz power. HR is accepted only if SNR ≥ 0 dB; pure noise passes this less than 5 % of the time.

3 Spectral purity (SQI) $= 50.0\,\%$. Same quantity as Dashboard item 9; 50.0 is the clamped floor.

4 Respiration $f_r = 0.23$ Hz (14.1 br/min). Peak of the 0.15–0.40 Hz respiratory spectrum; $60 \times 0.235 = 14.1$.

5 "Wavelet ratio" $D_2/A_3 = 46.97$. Energy ratio $\sum D_2^2 / \sum A_3^2$. *Viva note:* it is **not** a true wavelet transform. A_3 is

a moving average of the BVP (about 1 s window), D_2 is BVP $- A_3$ and D_1 is the first difference. A 1 s average almost cancels a 1 Hz pulse, so A_3 is tiny and the ratio is large. The current build names this honestly: trend / pulsatile / difference.

6 **Video source.** Which camera feeds the engine (ESP32-S3 over its parallel DVP camera bus). "15 FPS" was a fixed label.

7 **rPPG power spectral density.** *What:* how the pulse signal's power is spread over frequency (0.5–3.5 Hz shown, in dB relative to the maximum). *How:* **Welch's method**: split the signal into 50 %-overlapping segments, apply a Hann window, take each FFT magnitude squared, and average. Averaging lowers the variance of the estimate. **Reading this plot:** the curve is almost *flat* from 0.9 to 2.3 Hz, with several bumps within 1–2 dB (0.94, 1.55, 1.9, 2.25 Hz). So the 0.94 Hz "winner" is barely ahead: a noisy recording. The straight segments come from coarse resolution: 128-sample segments at 15 Hz give bins $15/128 = 0.117$ Hz $= 7$ BPM apart. *Current build:* 4096-point zero-padding plus parabolic interpolation (sub-bin peak), peak tracking near the previous HR, and the SNR gate. The fall above 2.5 Hz is the band-pass filter edge.

8 **Acoustic STFT spectrogram.** *What:* time (x, s) versus frequency (y, kHz), colour = level in dB (white $= -70$ dB floor, dark green = loud). *How:* **short-time Fourier transform**: FFT of short successive audio blocks (Web Audio analyser, FFT 512), one column per update. The tooltip (8.2 s, 2.27 kHz, −70 dB) is a silent cell. **Reading:** energy sits mostly below 1.5 kHz (voice/room noise); vertical stripes (e.g. at 0 and 8 s) are broadband clicks or speech. Heart sounds (20–150 Hz) would occupy only the bottom one or two rows (each row is about 90 Hz wide), so the PCG is analysed separately with a 25–400 Hz filter and envelope autocorrelation (Dashboard item 12).

9 **Mean-centred normalisation.** The BVP in z-score units: $z = (x - \mu)/\sigma$. Removing the mean (DC) and scaling to unit SD makes signals from different people, lighting and cameras comparable and is required before CHROM/FFT. Peaks $\approx$ beats; the shrinking amplitude after 63 shows the signal fading (movement or light change).

10 **Multi-resolution decomposition.** Plots A_3 (slow trend, green), D_2 (pulsatile residual) and D_1 (first difference) from item 5. D_2 carries the heartbeat oscillation; A_3 near zero shows little slow drift. Same viva note as item 5: moving-average split, not a DWT.

4 Viva question bank

Short answers you can say out loud. **Bold** = the key point. Where a question is a trap, the answer says so.

A. Motivation and design choices

Q. Why not just use a phone or a webcam? The ESP32 image is poor.

Not for image quality (a phone is better): for (1) **one clock** for camera and mic, so fusion is time-aligned; (2) **sensor control**: we lock exposure, gain and white balance after 3 s, which phones and browser webcams keep changing; (3) a **chest microphone** for heart sounds, which a phone filming your face can't provide; (4) a **cheap, standalone node** with its own display and Wi-Fi. Trade-off accepted: 240×160, about 8 fps. The same software also runs on a webcam (fallback mode), so a direct comparison is possible.

Q. What problem does the project solve?

Measuring vital signs **without touching** the person (no finger clip or chest strap): useful for burns, neonates, infection control and continuous monitoring. We add a second modality (heart sounds) to check the camera estimate.

Q. What is novel here?

Not the algorithms (CHROM, Welch, PCG envelopes are published). The contribution is the **integration**: a low-cost ESP32 node that streams time-synchronised video and audio, a camera/acoustic **fusion with an agreement gate**, **quality gating** that shows "–" instead of noise, and an honest study of what such a cheap sensor can and cannot do.

Q. Why "SonicVitals"?

"Sonic" for the acoustic (heart-sound) channel and "Vitals" for HR/RR; the system is multimodal: optical plus acoustic.

Q. Who would use it?

A research or screening prototype: telehealth kiosks, hostels and labs, low-resource clinics. **Not a medical device**; it would need clinical validation and regulatory approval.

B. Hardware

Q. Why the ESP32-S3?

Dual-core 240 MHz Xtensa LX7, **8 MB PSRAM** (frame buffer), a camera (DVP) interface, an **I²S** interface for the mic, built-in **Wi-Fi**, low cost. Two cores let audio and camera run in parallel without blocking.

Q. Which camera, and what is its limitation?

GC2145 (2 MP CMOS). It has **no hardware JPEG encoder**: the driver rejected JPEG mode, so we capture RGB565 and compress in software (quality 60, about 5 kB per frame). That limits us to about 8 fps over Wi-Fi.

Q. Why the INMP441 microphone?

A digital I²S MEMS mic: 24-bit output, so no analogue amplifier or ADC noise, and it is cheap and small. **Limitation:** its response falls off below about 60 Hz, while heart sounds are 20–150 Hz, so it must be pressed on the chest; in air the heart sounds are weak.

Q. What is the OLED for?

An SSD1306 128×64 display over I²C (GPIO47 SDA / 48 SCL, address 0x3C). It shows HR/RR/status sent back by the host and the node's IP address, so the device is usable without the dashboard.

Q. Why couldn't you use USB for everything?

The board has one USB port, wired to a CH340 USB-to-serial chip on **GPIO43/44**. The mic's I²S clock lines were on the same pins, so with the mic on, the serial link died (verified). Also the CH340 was unreliable above 1 Mbaud. So we stream over **Wi-Fi**; USB is only for flashing and power.

Q. How is it powered? Power consumption?

5 V over USB (laptop or power bank). **Power was not yet measured**; planned with a USB power meter (Table III of the paper). Do not quote a number you haven't measured.

Q. Why PSRAM?

A frame ring buffer (24 slots × 16 kB ≈ 3 s) absorbs Wi-Fi stalls, so frames are not lost when the hotspot briefly slows down.

C. Firmware and communication

Q. How is the firmware organised?

FreeRTOS tasks: **audio task on core 0** (reads I²S DMA, filters, queues blocks), **camera task on core 1** (captures at a fixed 8 fps, JPEG-encodes, puts frames in the PSRAM ring), and the main loop sends data, reads host commands and updates the OLED.

Q. How are camera and audio synchronised?

Every frame and audio block carries a **microsecond timestamp from the same ESP32 clock** (`esp_timer`) plus a sequence number. The host analyses by timestamp, not arrival time, so network delay doesn't matter.

Q. Why TCP and not UDP?

A lost UDP packet would corrupt a JPEG frame. TCP guarantees order and delivery. Latency isn't critical because data is timestamped at capture.

Q. Why not Bluetooth?

BLE throughput (tens of kB/s in practice) is too low for video plus audio; Wi-Fi gives about 70 kB/s on a phone hotspot, which is enough.

Q. What does the protocol look like?

Text lines: `«HELLO:...»`, `«FRAME_START:seq,t_us,len»` + base64 JPEG + `«FRAME_END»`, `«AUDIO:seq,t_us,n,data»`, `«STAT:...»` (fps, drops). Downlink `«VITALS:HR,RR,SpO2,Q,STATUS»` to the OLED. The length field and sequence numbers detect corrupt or missing data.

Q. Why base64, which adds 33 %?

It keeps the stream as plain text lines, so the parser can always re-synchronise at the next « marker and it is easy to debug. The cost is 4/3 bandwidth; binary framing is future work.

Q. Why send JPEG and not raw pixels?

Raw: 240 × 160 × 2 = 76.8 kB per frame, 614 kB/s at 8 fps, far above the link. JPEG is about 5 kB. Downside: compression removes some of the tiny colour changes (McDuff et al., 2017).

Q. Why decimate audio from 16 kHz to 4 kHz?

Bandwidth: 16 kHz × 2 bytes = 32 kB/s (43 kB/s in base64) would starve the video. Heart sounds are below 400 Hz, so 4 kHz (Nyquist 2 kHz) keeps them. A **63-tap FIR low-pass** filter runs first to prevent **aliasing**.

Q. Why lock exposure, gain and white balance?

Auto-exposure changes brightness every few frames, a change much larger than the pulse (0.1–1 %). After 3 s of settling, we freeze them so colour changes come from the skin, not the camera.

Q. How does the laptop find the board?

The node joins the phone hotspot and announces `sonicvitals.local` (mDNS); the host also scans the /24 subnet for port 3333. If no network is configured, the node becomes an access point (`SonicVitals-Node`, 192.168.4.1).

Q. How did you verify the link quality?

Measured over 30 s: **8.0 fps**, frame-timing jitter **0.4 ms**, **0** dropped or corrupt frames, **0** audio blocks lost (sequence numbers prove it).

D. rPPG theory and signal processing

Q. What is photoplethysmography?

Measuring blood-volume changes optically. In a pulse oximeter a LED shines *into* the finger; **remote** PPG uses ambient light and a camera at a distance.

Q. Why does skin colour change with the heartbeat?

Each beat pushes blood into the capillaries; haemoglobin **absorbs green light strongly** (less red), so reflected green dips slightly in systole. The change is about 0.1–1 % of the brightness.

Q. How can a change below 1 grey level be measured?

By **averaging thousands of pixels**: random noise falls as $1/\sqrt{N}$,

while the pulse is common to all skin pixels and survives.

Q. Why the forehead and cheeks?

Thin skin, dense capillaries, little hair, and relatively still. Forehead plus both cheeks gives more pixels than the forehead alone, so lower noise. Beard and eyebrows are avoided.

Q. Explain CHROM.

Normalise R, G, B by their means (removes light intensity). Build $X = 3R_n - 2G_n$ and $Y = 1.5R_n + G_n - 1.5B_n$; both contain the motion/specular part, so $S = X - \alpha Y$ with $\alpha = \sigma_X/\sigma_Y$ **cancels it** and keeps the pulse (de Haan & Jeanne 2013).

Q. Why not just the green channel?

Green has the strongest pulse but also all the lighting and motion changes. CHROM is more robust; we keep green as a baseline for comparison (ablation).

Q. CHROM vs POS vs deep learning?

POS (Wang 2017) is a similar projection; either works. Deep models (DeepPhys, PhysNet) can be more accurate but need large training sets and a GPU, and may not generalise to a new low-quality camera. CHROM needs no training.

Q. Why band-pass 0.75–2.5 Hz?

0.75–2.5 Hz = 45–150 BPM, the expected range. It removes breathing (< 0.4 Hz) and high-frequency noise. **Limitation**: HR below 45 (athletes) would be missed.

Q. Why Butterworth, and why filtfilt?

Butterworth has a **maximally flat passband** (no ripple distorting amplitudes). `filtfilt` runs it forwards and backwards, giving **zero phase shift**, so peaks are not delayed.

Q. Why resample to 15 Hz?

Frames arrive at uneven times over Wi-Fi; the FFT needs uniform samples. Each frame is placed at its capture timestamp and linearly interpolated onto a uniform grid.

Q. Is 8 fps enough? (Nyquist)

The sampling rate must exceed twice the highest frequency. HR up to 150 BPM is 2.5 Hz, which needs > 5 fps; 8 fps gives 4 Hz Nyquist. Enough for HR, not for fine waveform shape or HRV.

Q. What is Welch's method and why use it?

Split the signal into 50 %-overlapping segments, apply a Hann window, average the periodograms. Averaging **reduces the variance** (noise) of the spectrum compared with a single FFT.

Q. Why a Hann window?

Cutting a signal abruptly causes **spectral leakage** (power smeared into other bins). The Hann window tapers the ends and reduces leakage.

Q. What is the frequency resolution?

$\Delta f = 1/T$. A 10 s window gives 0.1 Hz = 6 BPM. **Zero-padding** (FFT length 4096) only interpolates the spectrum; **parabolic interpolation** through the peak and its neighbours then estimates the peak between bins.

Q. Why wait 10 s before showing HR?

To get about 8–15 beats and a usable frequency resolution. Shorter windows give unstable peaks. The analysis buffer is 30 s.

Q. How do you know a reading is good? (SNR)

The de Haan **SNR**: power near f_0 and its harmonic $2f_0$ divided by the rest of 0.5–4 Hz, in dB. HR is accepted only if SNR $\geq$ 0 dB. We set this threshold by testing pure noise: it passes in fewer than 5 % of cases.

Q. What is spectral concentration Q?

The share of in-band power within ±0.1 Hz of the peak. High Q means one clear peak. It is used as the camera's fusion weight.

Q. What happens when the person moves briefly?

Gaps up to **2 s** are bridged by interpolation, so the estimate continues. Longer breaks restart the 10 s buffer while the last value is **held** (marked "HELD") for up to 15 s.

Q. How do you stop the HR jumping between peaks?

Peak tracking: if a peak within ±12 BPM of the last accepted HR has at least half the power of the highest peak, it is kept. The displayed value is a median of the last 3 estimates followed by an exponential moving average (EMA).

Q. How does face detection work?

Viola–Jones Haar cascade: Haar-like rectangle features computed quickly from an integral image, chosen by AdaBoost, arranged in a cascade that rejects non-faces early. Fast on a CPU. The box is smoothed to reduce jitter.

Q. Why not a deep face detector (MediaPipe)?

Haar is light and real-time on a laptop CPU. MediaPipe Face Mesh would track landmarks better under head turns: **future work**.

Q. What are the five face-gate tests?

Not touching the edges; size 20–88 % of the frame width; aspect ratio 0.65–1.30 (not turned); all ROIs inside the frame; enough light and similar left/right brightness. Frames that fail are not used.

Q. Two faces in the frame?

The largest face is used. Multi-person tracking is future work.

E. Respiration, SpO_2 and HRV

Q. How do you get the breathing rate from a face?

Breathing modulates the skin signal (venous return, chest and head movement, light angle): **respiratory-induced intensity variation**. Band-pass green 0.15–0.40 Hz (9–24 breaths/min), take the spectral peak. Needs 15 s and a clear peak ($\geq$ 50 % of the low-frequency power), otherwise "–".

Q. Can a camera measure SpO_2?

Only roughly. Pulse oximeters use the **ratio of ratios** $(AC/DC)_{red}/(AC/DC)_{IR}$; our camera has no infrared, so we use red/blue as a proxy. It needs **calibration** against a reference oximeter ($SpO_2 = A - B \cdot R$). Without that file, the dashboard shows only the uncalibrated ratio.

Q. What is HRV and how do you measure it?

Beat-to-beat variation of the inter-beat interval (IBI). RMSSD $= \sqrt{\text{mean}(\Delta IBI^2)}$. We compute it from heart-sound beat times and need $\geq$ 20 clean intervals. At 8 fps the camera's time resolution (125 ms) is too coarse for HRV.

Q. What is LF/HF?

The ratio of HRV power at 0.04–0.15 Hz to 0.15–0.4 Hz, used as a sympathovagal-balance (stress) index. It needs several minutes of accurate beats, so we **removed** the stress card (it was a placeholder).

F. Heart sounds (PCG)

Q. What are S1 and S2?

S1 ("lub"): mitral and tricuspid valves closing at the start of systole. S2 ("dub"): aortic and pulmonary valves closing at the end. Main energy is about 20–150 Hz.

Q. How do you get HR from sound?

Band-pass 25–400 Hz, resample to 1 kHz, **Shannon energy** envelope $-x^2 \log x^2$ smoothed over 20 ms, then **autocorrelation**: the first strong peak at lag τ (0.4–1.33 s) is one beat period, so HR $= 60/\tau$.

Q. Why Shannon energy and not just x^2?

x^2 is dominated by the loudest peaks. Shannon energy emphasises **medium-intensity** sounds and suppresses both low noise and very high spikes, giving a cleaner envelope.

Q. Why autocorrelation instead of peak picking?

It uses the **periodicity of the whole window**, so it is robust to a few missed or extra peaks; it does not need to label S1 vs S2.

Q. What is ρ?

The autocorrelation value at the beat lag (0–1): how periodic the sound is. Below 0.3, the acoustic HR is not reported.

Q. Did the heart-sound channel work?

Honestly: weakly. In air, 99.5 % of the audio power was below 25 Hz and ρ stayed $\leq$ 0.18, because the INMP441 cuts off around 60 Hz. It needs firm chest contact or a stethoscope-coupled mic:

future work.

G. Fusion

Q. How do you fuse camera and sound?

Quality-weighted mean $(w_o HR_o + w_a HR_a)/(w_o + w_a)$, with $w_o = Q$ (camera) and $w_a = \rho$ (sound). **Agreement gate:** only if they differ by ≤ 8 BPM; otherwise use the better channel and flag "CONFLICT".

Q. Why an agreement gate?

Averaging a correct value with a wrong one gives a wrong answer. If two independent sensors disagree, one is failing, so we don't average.

Q. Why 8 BPM?

Slightly above the AAMI tolerance (± 5 BPM) to allow for each channel's own error. It is a tunable parameter, to be re-tuned on development data.

H. Colour indices (Diagnostics Studio)

Q. Can a camera detect jaundice or anaemia?

It can give **screening hints**. Research (e.g. BiliScreen) used colour-calibration cards and controlled light. Ours are **uncalibrated and exploratory**; lighting and skin tone change them a lot.

Q. Your screenshot says jaundice ALERT. Do you have jaundice?

Trap. No: warm room light lowers blue, and the eye band includes skin and eyebrows, which raises the yellowness index. It shows exactly why calibration is needed.

Q. Why log in the erythema index?

By the Beer–Lambert law, absorbance is logarithmic in reflectance, so $\log R - \log G$ tracks haemoglobin absorption roughly linearly.

Q. What is the Laplacian variance?

A sharpness measure: the variance of the 2nd-derivative image. Blurred frames have weak edges and a low value; we pick the sharpest frame for colour analysis.

Q. How would you make colour indices reliable?

A colour reference card in the frame, fixed lighting, white-balance correction, and validation against lab tests (bilirubin, haemoglobin) with ethics approval.

I. Validation, accuracy and limitations

Q. How accurate is your heart rate?

Not yet validated against a reference device; we will not claim a number we haven't measured. Validated so far: the algorithms on synthetic signals (31 automated tests), and the streaming link (8 fps, 0 loss). Planned: ECG chest strap as reference.

Q. How will you validate it?

Simultaneous ECG chest strap (HR, R–R intervals), finger oximeter, metronome breathing; conditions at rest, talking, small motion, low light, different distances; 30 s windows; **MAE, RMSE, Pearson** r, **Bland–Altman** bias and 95 % limits of agreement, and the **AAMI EC13** criterion (error ≤ 5 BPM or 10 %).

Q. What is a Bland–Altman plot?

Difference (device − reference) against their mean. The **bias** is the mean difference; the **limits of agreement** are bias ± 1.96 SD. It shows agreement; correlation only shows that two values move together. With several windows per person, the repeated-measures version is used.

Q. Why not just use correlation?

Two methods can correlate perfectly yet differ by 10 BPM. Agreement needs bias and limits (Bland & Altman, 1986).

Q. Why split development and test participants?

Thresholds (SNR gate, Δ, RR gate) are tuned on development participants only, so test results are not biased by tuning.

Q. Main limitations?

Motion and **lighting changes**; low resolution, 8 fps and JPEG compression; **skin tone** (more melanin means a weaker pulse signal, a known bias in rPPG); beard and hair covering ROIs; weak heart sounds through air; no clinical validation yet.

Q. Does skin tone matter?

Yes. Melanin absorbs light and reduces the pulse amplitude, so SNR is lower for darker skin (Nowara et al., 2020). The SNR gate avoids wrong numbers but may show "–" more often. Diverse participants are needed.

Q. What about fluorescent light flicker?

Mains lighting flickers at 100 Hz. With 8 fps sampling it can alias into the low-frequency band as a fake peak. Locking the exposure and using daylight or DC LED light reduce it.

Q. What if I show it a photo of a face?

The face detector will find it, but a photo has no pulse: the spectrum is noise, the SNR stays below the gate, and HR shows "–". (A photo under flickering light could still pass occasionally; liveness detection is future work.)

Q. Why did your earlier screenshot show 97 BPM and another 57 BPM?

Different moments; and in that first build there was no quality gate, so noisy windows were still displayed. The current build only shows estimates that pass the SNR gate.

Q. Your HRV was exactly 45.0 and stress exactly 50.0. Why?

Trap; be honest. In the first build those were placeholder formulas driven by the quality score; with quality at its floor (50) they gave 45.0 and 50.0. I found this, removed them, and replaced HRV with real RMSSD from heart sounds.

Q. What is the latency?

About 10 s for the first estimate (buffer), then an update every 2 s on the dashboard and every 1.5 s to the OLED.

J. Software

Q. Software stack?

Firmware: Arduino core on ESP-IDF/FreeRTOS, built with PlatformIO. Host: Python, Flask, OpenCV, NumPy/SciPy. Frontend: HTML/CSS/JavaScript with Plotly charts, polling the Flask JSON API.

Q. Why process on the laptop instead of on the ESP32?

Face detection, CHROM and spectral analysis in Python are easy to develop, debug and change. The ESP32 could run a simpler version (e.g. fixed ROI plus FFT): **edge processing** is future work and would also remove the video stream.

Q. How do you record data for analysis?

The "Experiments" tab starts a session: `frames.csv` (timestamps, ROI colour, gate), `audio.wav`, `estimates.csv`, `meta.json`. `analysis/evaluate.py` replays them offline with the **same** DSP code.

Q. How did you test the software?

31 automated `pytest` tests with synthetic signals of known HR/RR (e.g. 52/78/128 BPM within 2 BPM), protocol parsing, gap bridging, noise rejection, and a node simulator that imitates the ESP32 stream.

Q. Privacy and security?

Everything stays on the local network; nothing goes to the cloud; video is saved only during an explicit recording. Limitation: the TCP stream is not encrypted (WPA2 only protects the Wi-Fi link).

K. Future work

Q. What would you improve first?

(1) A validation study with an ECG reference; (2) compute ROI colour means **on the ESP32 from raw pixels** before JPEG (avoids compression loss); (3) a stethoscope-coupled or contact mic; (4) a landmark face tracker; (5) an IR camera/LED for SpO_2; (6) edge processing and a battery.

Q. Could this become a product?

After clinical validation, IEC 60601 electrical-safety testing and regulatory approval (CDSCO in India) as a medical device. Today it is a research prototype.